

Technology Stack

Date	28 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>HTML, CSS, JavaScript</i>	<i>Used to create a simple and responsive user interface for entering HDI indicators and displaying prediction results.</i>

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	<i>Python (Flask)</i>	<i>Handles user requests, processes input data, and communicates with the machine learning model.</i>
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>CSV Dataset</i>	<i>CSV file is used for training, while the saved model (.pkl) stores the trained machine learning model for prediction.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>Local Flask Server</i>	<i>Used to run the application locally during development and testing.</i>

5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	<i>GitHub</i>	<i>Used to manage source code, track changes, and collaborate during development.</i>
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Scikit-learn, Pandas, NumPy</i>	<i>Scikit-learn trains the prediction model, Pandas handles data processing, NumPy performs numerical operations, and Joblib saves and loads the trained model.</i>